

RFID-Based Door Lock System Using Arduino

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Abstract—The objective of this paper is to present the design and development of a smart door lock system based on RFID technology, utilizing an Arduino microcontroller as the main controller for secure and automated access control systems. The system uses an RC522 RFID module to identify and recognize the unique identification (UID) of RFID cards/tags. The Arduino microcontroller then processes this information and compares it with pre-set authorized IDs to grant access permission to the user. Once access is granted, a relay module is activated to switch a solenoid lock, thus granting temporary access to the user, while a 16x2 I2C LCD displays real-time information to the user regarding access permission. The proposed system provides an efficient, cost-effective, and reliable security solution for access control in various applications, replacing conventional key-based systems with an RFID-based access control system for secure access to restricted areas.

Keywords: *Arduino, RFID, AccessControl, DoorLockSystem*

I. INTRODUCTION

The necessity for security in residential, business, and industrial sectors has prompted the creation of advanced access control systems. The limitations of key-based locking systems include the need to duplicate keys, the risk of losing keys, and the absence of access control. These limitations of key-based locking systems demonstrate the need for a more secure and efficient locking system.

RFID (Radio Frequency Identification) technology is a highly reliable and contactless technology for access control systems. Access control systems based on RFID technology are commonly used for security systems, attendance management systems, toll collection systems, and identification systems. Using RFID technology for access control systems ensures that access to the system is provided to authorized individuals only. This improves the security of the access control system.[1]

Significant advancements have been made in the field of embedded systems and microcontrollers. This has led to the creation of cost-effective and efficient smart security systems. Arduino technology is one of the most efficient and cost-effective microcontrollers. Using RFID technology and Arduino technology, it is possible to design a door locking system.

This project is mainly concerned with the design and implementation of a smart door lock system that uses RFID technology and is controlled by an Arduino microcontroller. The system reads the UID of the RFID card and compares it with the stored authorized IDs. Once it finds a matching ID, it opens the door lock using a solenoid lock and a relay

module. A 16x2 I2C LCD display is used to display real-time information to the user.[3]

A. Overview of RFID-Based Access Control Systems

RFID-based access control systems use radio frequency signals to identify and authenticate users. These systems typically consist of three main components:

- RFID reader module
- RFID cards or tags
- Microcontroller-based processing unit

The RFID reader sends radio signals to identify the presence of RFID tags. When the RFID tag is brought close to the RFID reader, it sends its unique identification number. This identification number is then processed by the microcontroller to decide whether access is to be granted or denied.

RFID technology is commonly used due to its keyless operation, fast response time, and reliability. It does not require the use of keys and minimizes the risk of unauthorized access.

B. Working Principle of RFID Door Lock Systems

In the case of the RFID door lock system, the RFID reader reads the card or tag, sending the UID to the microcontroller. The microcontroller then compares the scanned UID of the card or tag with the list of predefined IDs stored in the memory.

If the UID of the scanned card or tag matches the list of IDs, the system opens the door by activating the relay module, which operates the solenoid door lock. However, if the UID does not match the list of IDs, the system denies the request, keeping the door locked.[3]

C. Aim of the Project

The aim of this project is to design and develop a smart RFID-based door lock system using Arduino to provide a secure, reliable, and contactless access control solution.

D. Objectives of the Project

The main objectives of this project are:

- To design an RFID-based access control system using Arduino
- To implement UID-based authentication for security
- To control a solenoid door lock using a relay module
- To display system status using an LCD display
- To develop a low-cost and user-friendly security system

E. Scope of the Project

The project will be based on the design, implementation, and testing of a small-scale door locking system based on RFID technology. The project can be applied in homes, offices, and other organizations.

The project will include hardware integration, programming, and testing of the implemented functionality. Other advanced functionalities such as remote monitoring and the use of biometrics will be excluded and may be considered for future enhancements.[1]

II. LITERATURE REVIEW

A. RFID Technology in Access Control Systems

The RFID technology, which stands for Radio Frequency Identification, is used in modern access control systems owing to its non-contact operation, high-speed response, and reliability. The RFID system makes use of electromagnetic fields to identify and track tags attached to objects or carried by individuals. The RFID system finds widespread use in security systems, attendance tracking, toll booths, and inventory control, amongst many other applications [3].

As per Finkenzeller [3], the RFID technology makes it possible to provide identification and access control without physical contact, which makes it suitable for use in door access control systems. The RFID technology is considered to be much more convenient and secure, with fewer chances of unauthorized reproduction, as compared to traditional systems based on keys. However, the limitations and security issues associated with the system must also be taken into consideration.

The RFID technology will be used in this project to design a secure door lock system, allowing access only to authorized personnel based on their unique identification.

B. Microcontroller-Based Embedded Security Systems

One of the important components of embedded security systems is the use of microcontrollers. The Arduino-based system has gained popularity because of its ease of use and hardware integration. The Arduino-based system can easily interface various sensors and communication modules. Therefore, it is suitable for security system implementation [5]. The Arduino-based system is used to control access systems. The Arduino-based system can easily process the input data and make decisions for security system implementation. The use of microcontrollers may have limitations in processing large-scale applications. However, Arduino is suitable for small to medium-scale security system implementation, such as RFID-based door locks.[1]

C. Relay-Controlled Electronic Locking Mechanisms

Electronic door lock systems typically employ relays for controlling high-power devices such as solenoid door locks. A relay is essentially a switch that allows for control of a device at a higher voltage using a lower voltage signal from a microcontroller.

Solenoid door locks are popular because they are easy to implement and are quite reliable. When energized, the solenoid retracts, causing the door to unlock. When the solenoid is switched off, it returns to its default position, causing the door to lock.[4]

According to Rouse [2], relay control systems offer safety advantages in terms of electrical isolation. However, switching relays does cause slight delays and wear over time. Nevertheless, this is quite acceptable for door lock control systems.

D. User Interface and Display Systems in Embedded Applications

The user interface components, such as LCD displays, play a vital role in embedded systems. A 16x2 I2C LCD display is commonly used in embedded systems, as it is simple and requires fewer connections.

The I2C communication protocol is used to communicate between devices, and it requires only two wires to connect multiple devices. The LCD displays can be used to display system messages, such as access granted, access denied, and system status.[5]

The user interaction with the system can be improved by providing proper feedback. The LCD displays have limited capabilities, and they can only be used to display text-based information.

E. Security Challenges in RFID-Based Systems

Though RFID systems have numerous advantages, there exist some security issues that need to be addressed. Unauthorized access, RFID card cloning, and signal interception are security issues that can arise in RFID-based access control systems [1].

However, security can be enhanced through various means such as encryption and the use of other security systems such as multi-factor authentication and the use of other technologies such as biometric systems. The system that is proposed in this project is mainly based on simple UID authentication. However, it can be improved to add security to overcome the security issues.

III. METHODOLOGY

A. Overall System Architecture

The proposed RFID door lock system consists of four main components:

- RFID reader module
- Microcontroller (Arduino Uno)
- Relay-controlled solenoid lock
- User interface (LCD display)

Working principle:

- 1) RFID card is presented to the reader.
- 2) Reader sends UID to the microcontroller.
- 3) Microcontroller verifies the UID against stored authorized IDs.
- 4) Relay activates the solenoid to unlock the door if authorized; otherwise access is denied.
- 5) LCD shows "Access Granted" or "Access Denied".

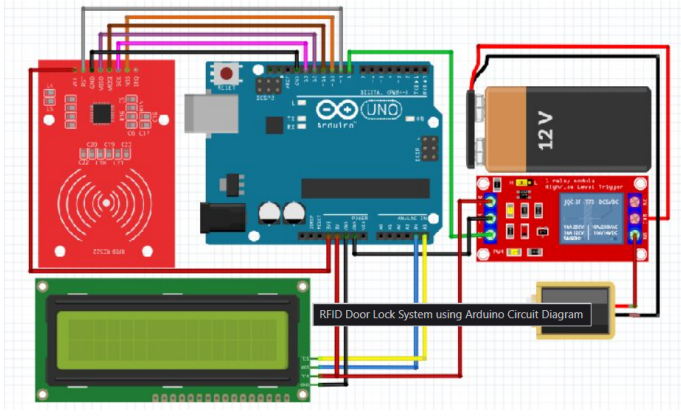


Fig. 1. Block diagram of RFID Door Lock system

B. RFID Reader Subsystem

The RFID reader detects and reads RFID cards and sends the UID to the microcontroller via a serial interface.

- Contactless operation for security and convenience.
- Fast read time (<100 ms).
- Compatible with 13.56 MHz RFID cards.

C. Microcontroller Subsystem

Arduino Uno acts as the main controller. It performs:

- UID verification
- Relay control to lock/unlock solenoid
- LCD display control

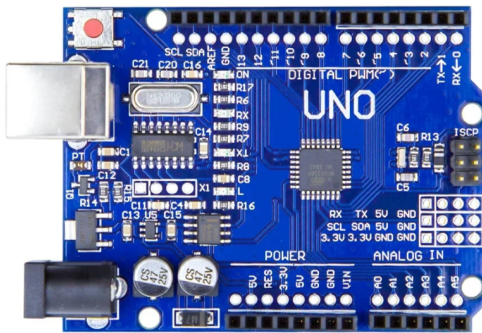


Fig. 2. Arduino Uno controlling the system

D. Solenoid Lock and Relay Subsystem

The solenoid lock is controlled by a relay to safely switch high current from the microcontroller.

- Relay provides electrical isolation
- Solenoid locks automatically when power is off
- Fast response for door access

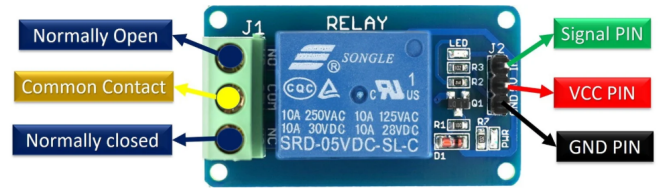


Fig. 3. Solenoid lock connected via relay to Arduino

E. User Interface Subsystem

The 16x2 LCD display provides feedback:

- “Access Granted”
- “Access Denied”
- System status messages

F. Security and Safety Measures

- Only authorized UIDs can unlock the door
- Relay and solenoid protect the microcontroller from high current
- Optional buzzer alerts unauthorized access attempts

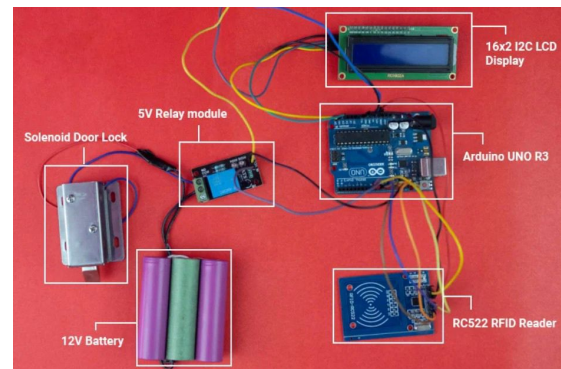


Fig. 4. Complete system highlighting safety and security features

Security is a major concern in modern homes and offices. RFID technology is widely used due to its contactless operation and fast response time [3]. In this project, an Arduino UNO is interfaced with an RC522 RFID reader to read the unique identification number (UID) of RFID cards and tags.

When an RFID card is placed near the reader, the system captures the UID and displays it on a 16x2 I2C LCD display as well as on the serial monitor. This UID can later be stored and used to verify authorized users [4].

IV. WORKING PRINCIPLE

The working steps of the system are as follows:

- 1) Power ON the system
- 2) Place RFID card near the reader
- 3) Arduino detects the card
- 4) UID is read and processed
- 5) UID displayed on LCD and Serial Monitor
- 6) System waits for next scan

- [4] NXP Semiconductors, “MFRC522 RFID Reader Datasheet,” 2019.
 [5] Arduino Documentation, “SPI Communication,” Available: <https://www.arduino.cc>

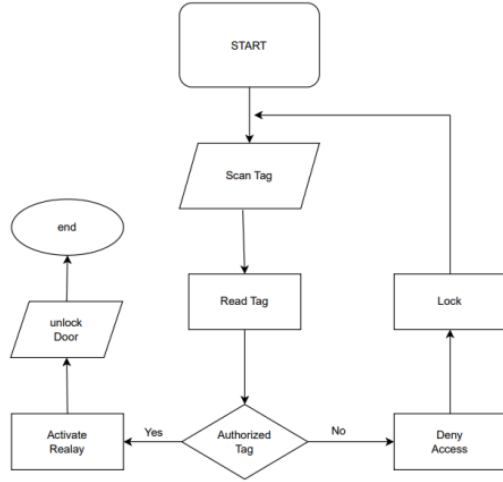


Fig. 5. flow chart

V. EXPECTED RESULTS

- Accurate detection of RFID cards
- Correct UID displayed on LCD
- Reliable serial output for debugging
- UID can be stored for door lock authorization
- Fast system response time.

Table 1 outlines the main functional specifications of the smart door lock system. The Door Unlocking function provides Lock/Unlock capability with single-step execution [1]. The RFID Authentication operates within 0-5 cm range with 0.1 cm resolution for accurate contactless card reading [4]. The Entry Logging maintains a closed database recording one entry per action for security tracking [2].

TABLE I
MAIN FUNCTIONS OF THE SMART DOOR LOCK

Function	Range / Type	Resolution / Detail
Door Unlocking	Lock / Unlock	1 step
RFID Authentication	0-5 cm	0.1 cm
Entry Logging	Cloud database	1 entry per action

The flowchart illustrates the operational workflow of the RFID-based door lock system. The process begins at the START node, followed by the "Scan Tag" operation where the RC522 RFID reader detects the card [4]. The system then reads the UID and evaluates it at the "Authorized Tag?" decision point [1]. If authorized, access is granted; if not, access is denied [2]. The system then loops back through the "Load" stage for continuous monitoring.

REFERENCES

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